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DS 5030

Lab 6

Diabetes Clinical Trial Analysis

A clinical trial was conducted considering three treatments for pre-diabetics to reduce the progression to full diabetes. The three treatments discussed in the study were Metformin, Rosiglitazone, and lifestyle adjustments (diet, exercise). After conducting my own analysis of the data from this trial, we will now walk through my findings and, based on those findings, I will give my unbiased review of the trial's results.

The dataset from the trial contains 699 observations, each representing a patient using one of the three treatments for pre-diabetics. This was a randomized controlled trial, so the treatments were assigned to patients at random. Of those patients, 232 used the Metformin treatment, 233 used the Rosiglitazone treatment, and 234 of them made lifestyle adjustments such as diet and exercise. For each patient, the trial found whether their treatment was successful or if it failed to prevent full diabetes. In our sample, we found the average success rate of Metformin, Rosiglitazone, and lifestyle adjustments were about 48.3%, 61.4%, and 53.4% respectively.

Our sample averages would lead one to believe that Rosiglitazone is the best of the pre-treatments and Metformin is the worst, as they have the highest and lowest average success rate. These were only the average outcomes for one sample, however. To provide more evidence on if these assumptions are true, I bootstrapped the average outcome for the three groups, as well as the three treatments effects, 10,000 times to approximate the density and distribution of each.

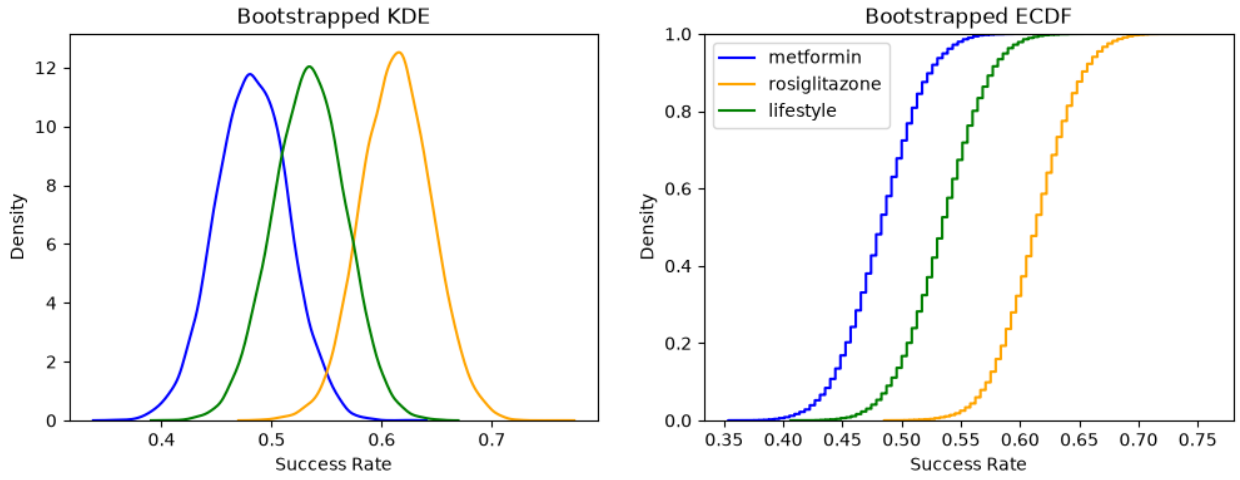


Figure 1: KDE and ECDF for the three diabetes treatments

The approximate ECDF and KDE for the average outcomes for each group from these samples can be seen in Figure 1. Looking at these plots, we see the curves for the Metformin treatment fall to the left of the curves for lifestyle and Rosiglitazone, while the curves for the Rosiglitazone treatment fall to the right of the others. This means in our estimates, there are more occurrences of lower success rates in the Metformin group than in the other two. In contrast, the Rosiglitazone treatment appears to be quite effective compared to metformin and lifestyle, as we estimated higher success rates in that group much more often. In fact, the ECDF appears to show that very few, if any, of the Rosiglitazone samples had an average success rate lower than 50%.

I also calculated the three treatment effects in each sample. The densities and distributions of these estimates are shown in figures 2, 3, and 4.

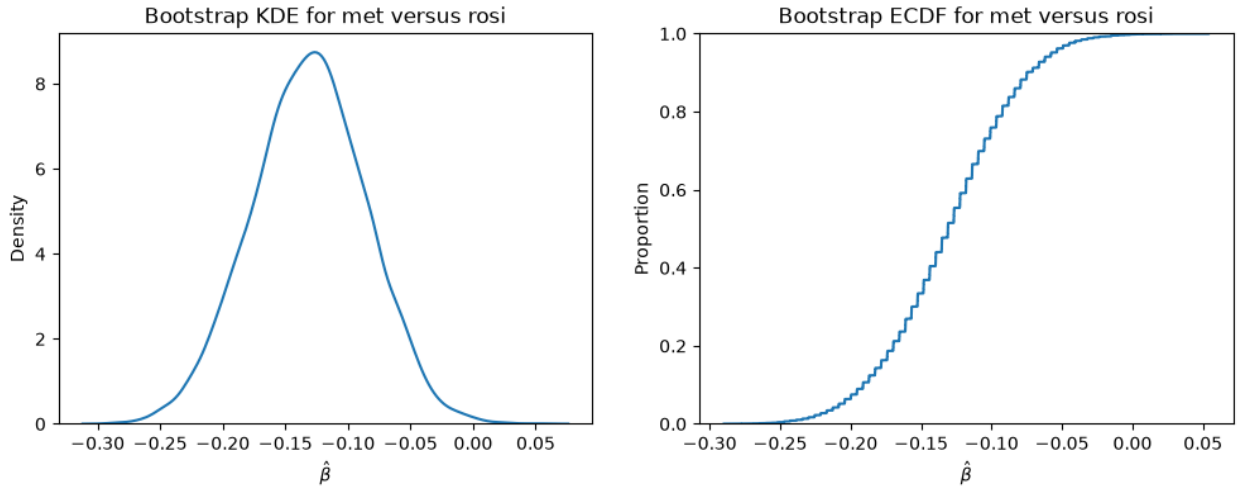


Figure 2: KDE and ECDF for the ATE of Metformin and Rosiglitazone

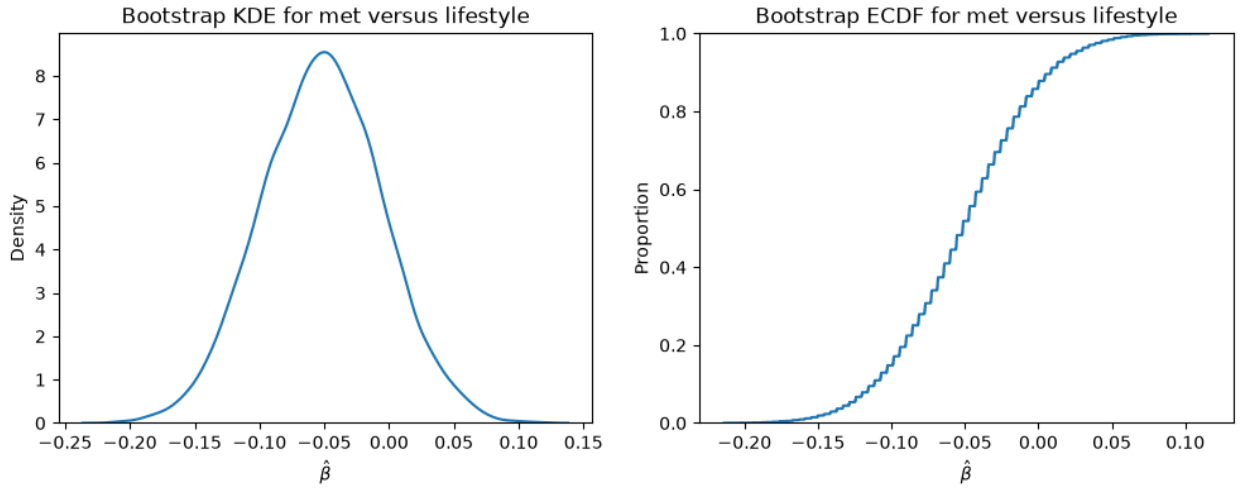


Figure 3: KDE and ECDF for the ATE of Metformin and Lifestyle Adjustments

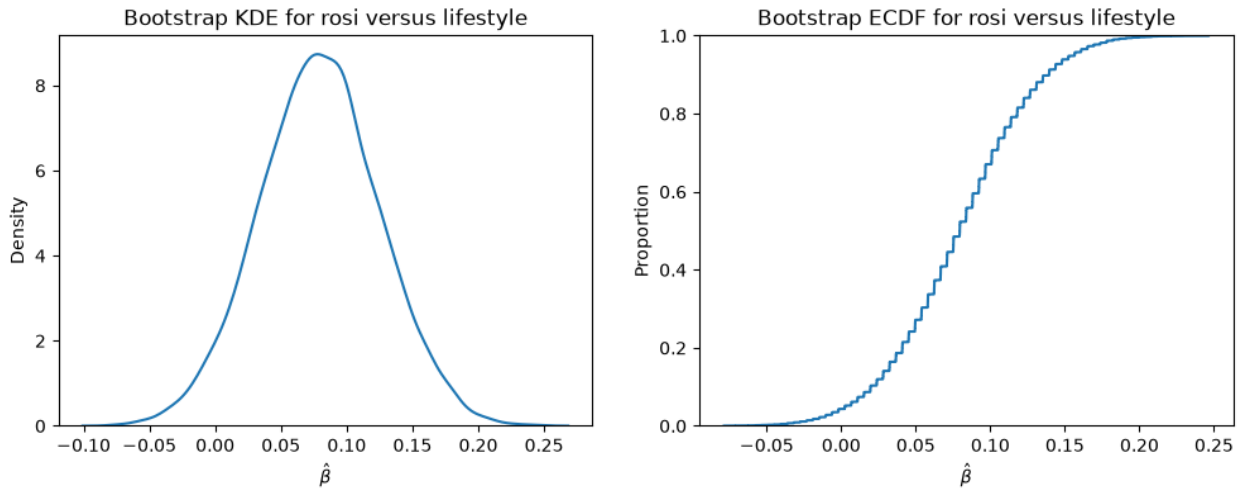


Figure 4: KDE and ECDF for the ATE of Rosiglitazone and Lifestyle Adjustments

This further emphasizes what we found in the distributions for the average outcomes. On average, our samples found negative ATE values when comparing the difference in outcomes for Metformin with those of Rosiglitazone and lifestyle adjustments. Rosiglitazone, however, seems to have a positive ATE on average when compared to lifestyle adjustments. To further illustrate this, I computed 90%, 95%, and 99% confidence intervals for each ATE based on its ECDF. The computed confidence intervals were as follows:

90% CI for met versus rosi: [-0.2083025, -0.05777342]

95% CI for met versus rosi: [-0.22130753, -0.04486042]

99% CI for met versus rosi: [-0.24722519, -0.01472538]

90% CI for met versus lifestyle: [-0.12864721, 0.02571471]

95% CI for met versus lifestyle: [-0.14168877, 0.03886678]

99% CI for met versus lifestyle: [-0.17167717, 0.06465517]

90% CI for rosi versus lifestyle: [0.00253109, 0.15654323]

95% CI for rosi versus lifestyle: [-0.01038159, 0.16945499]

99% CI for rosi versus lifestyle: [-0.03618732, 0.19529749]

The difference between Metformin and Rosiglitazone is significant, as 0 is not included in the 90%, 95%, or 99% confidence intervals calculated. All values in these intervals are negative, meaning Metformin performs worse on average than Rosiglitazone. The differences in Metformin and lifestyle adjustments do not appear significant, as 0 falls in all three intervals. As for the difference between Rosiglitazone and lifestyle adjustments, 0 does not fall in the 90% confidence interval, but does fall in the 95% and 99% confidence intervals. This means the

difference is significant at the 10% significance level, but not at the 5% or 10% significance levels. At the 10% level, however the difference between Rosiglitazone and lifestyle all ATE values in the interval are positive.

Based on my results, there is some evidence suggesting that Rosiglitazone does have a positive impact on preventing full diabetes. As seen in the ECDF of our bootstrapped samples, Rosiglitazone had very few success rates estimated below 50%. Its ATE values also showed it to be better than Metformin. For lifestyle adjustment, though, it the difference only appeared to be significant at larger significance levels. What is clear, however, is we find no evidence in my results to suggest Metformin helps prevent full diabetes. In fact, it may make it more likely to occur. The ECDF and KDE of our bootstrapped samples show lower success rates for Metformin compared to the other two groups. The difference between Metformin and Rosiglitazone is significant at all levels, with the intervals being comprised of only negative ATE values. We also did not find any significance in the difference between Metformin and lifestyle adjustments. This means Metformin performs worse than Rosiglitazone on average and shows no evidence of improvement over lifestyle adjustments.